

- 7 (a) curve levelling out (at $1.4 \mu\text{g}$) **M1**
 correct shape judged by masses at $nT_{1/2}$ **A1** [2]
 [for second mark, values must be marked on y-axis]
- (b) (i) $N_0 = (1.4 \times 10^{-6} \times 6.02 \times 10^{23})/56$ **C1**
 $= 1.5 \times 10^{16}$ **A1** [2]
- (ii) $A = \lambda N$ **C1**
 $\lambda = \ln 2 / (2.6 \times 3600) (= 7.4 \times 10^{-5} \text{ s}^{-1})$ **C1**
 $A = 1.11 \times 10^{12} \text{ Bq}$ **A1** [3]
- (c) 1/10 of original mass of Manganese remains **C1**
 $0.10 = \exp(-\ln 2 \times t/2.6)$
 $t = 8.63 \text{ hours}$ **A1** [2]
 [use of 1/9, giving answer 8.24 hrs scores 1 mark]

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